

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

<i>Q. No</i>	<i>Understanding of Concepts (25)</i>	<i>Experimental Design (30)</i>	<i>Use of Tools (20)</i>	<i>Analysis of Results (15)</i>	<i>Documentation (10)</i>	<i>Total Score (out of 100)</i>
<i>1</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>2</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>3</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>4</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>5</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>OVERALL RUBRICS VALUE</i>						
	<i>/ 5</i>	<i>/ 5</i>	<i>/ 4</i>	<i>/ 4</i>	<i>/ 2</i>	
	<i>/ 25</i>	<i>/ 30</i>	<i>/ 20</i>	<i>/ 15</i>	<i>/ 10</i>	<i>/ 100</i>

APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption. Deduce the most likely original message. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality. Demonstrate how the encryption process transforms the message step-by-step. Explain how the repeating keyword influences each character of the ciphertext.

Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection. Analyze how investigators could detect the presence of hidden data within the image. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

SIMATS ENGINEERING

ASSESSMENT TOOL 1

APPLICATION- BASED QUESTIONS

Name: Harshini R N

Reg No: 192421175

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5117

Scenario 1: Event Communication Security

- **Question:** A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.
- **Answer:**
 - **Encrypted Message Reconstruction:** Using a Caesar cipher with a shift of +4 (shifting each letter forward by 4 positions in the alphabet), the message "MEET AT NOON" transforms as follows:
 - M -> Q, E -> I, E -> I, T -> X
 - A -> E, T -> X
 - N -> R, O -> S, O -> S, N -> R
 - **Result:** QIIC EX RSSR
 - **Retrieval Process:** An authorized member aware of the encoding rule would decrypt the ciphertext by applying a reverse shift of 4 positions (-4) to each character (e.g., shifting Q back by 4 gives M), successfully recovering the original plaintext.

Scenario 2: Intercepted Suspicious Message

- **Question:** A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption. Deduce the most likely original message. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

- **Answer:**

- **Original Message Deduction:** The message is encoded using the **Atbash cipher** (a substitution cipher where each letter is mapped to its reverse in the alphabet: A<->Z, B<->Y, etc.). Applying this substitution yields:

- GSRH -> THIS

- RH -> IS

- Z -> A

- HVXIVG -> SECRET

- **Result:** THIS IS A SECRET
- **Justification:** Analyzing single-letter words and letter structures reveals that the isolated letter "Z" corresponds to "A" or "I". Testing classic monoalphabetic substitution ciphers shows that Atbash perfectly maps the ciphertext characters to coherent English words (e.g., G maps to T, S maps to H, R maps to I).

Scenario 3: Secure Messaging System

- **Question:** A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality. Demonstrate how the encryption process transforms the message step-by-step. Explain how the repeating keyword influences each character of the ciphertext.

- **Answer:**

- **Step-by-Step Encryption (Vigenère Cipher):**

- Plaintext: H E L L O (Numerical values with A=0: 7, 4, 11, 11, 14)
- Keyword (repeated): K E Y K E (Numerical values: 10, 4, 24, 10, 4)
- Calculation (Plaintext + Keyword mod 26):
- $H(7) + K(10) = 17 \rightarrow R$
- $E(4) + E(4) = 8 \rightarrow I$
- $L(11) + Y(24) = 35 \bmod 26 = 9 \rightarrow J$
- $L(11) + K(10) = 21 \rightarrow V$
- $O(14) + E(4) = 18 \rightarrow S$

□ **Ciphertext:** RIJVS ○ **Influence of the Repeating Keyword:** The keyword repeats cyclically to match the length of the plaintext message. Each letter of the keyword acts as a dynamic shift value for its corresponding plaintext character, which prevents simple frequency analysis attacks by varying the shift amount across different positions.

Scenario 4: Suspicious Digital Asset Investigation

- **Question:** During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection. Analyze how investigators could detect the presence of hidden data within the image. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.
- **Answer:** ○ **Detection Analysis:** Investigators can detect steganography using:
 - **Least Significant Bit (LSB) Analysis:** Inspecting the lowest bit planes of image pixels to uncover anomalous randomness where hidden messages are typically embedded.
 - **Statistical / Chi-Square Analysis:** Examining color palettes and pixel frequency distributions to identify deviations from natural image characteristics.

- **Visual Residual Analysis:** Filtering out higher-order bits to expose concealed patterns.
- **Recommendations to Strengthen Security:**
 1. **Payload Encryption:** Encrypt the secret message using a strong cipher (such as AES) prior to embedding, ensuring that even if the data is extracted, it resembles random noise.
 2. **Adaptive Steganography:** Hide data selectively within complex or noisy regions of the image rather than uniform areas, masking statistical signatures and avoiding detection.

Scenario 5: Legacy Encryption in Corporate Communication

- **Question:** A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEFAOCAIVDEN". a. Determine the most probable original message by reversing the encryption process. b. Critically evaluate why such a method is vulnerable in modern communication systems. c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.
- **Answer:**
 - **a. Original Message:** Reversing a multi-level transposition process (such as a 3-rail fence or columnar transposition) reorganizes the 25 characters back into their original sequence, yielding the plaintext: WE ARE DISCOVERED RUN AT ONCE.
 - **b. Vulnerability Evaluation:** Transposition ciphers only rearrange the positions of characters without changing their actual identities. As a result, they preserve standard letter frequencies and language patterns, making them highly vulnerable to anagramming, brute-force attacks, and statistical frequency analysis.

- **c. Practical Enhancement:** Implement a modern cryptographic standard such as **AES (Advanced Encryption Standard)** or combine transposition with substitution to ensure both strong confusion and diffusion.